

TAIMUR HASSAN SARMAD

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EDUCATION

Lahore University of Management Sciences

Sep. 2019 – June 2023

Bachelor of Science in Computer Science (Major **CGPA: 3.90/4.00**)

Lahore, Pakistan

WORK EXPERIENCE

FullStack Software Engineer

June 2023 – Present

PosterMyWall

LLMs, LangGraph, Typescript, React.js, Fabric.js, PHP

- Architected a **Multi-AI-Agent** system and **RAG** pipeline via an extended NeuronAI framework, cutting workflow latency by 3-4s and driving 165k+ chats and 96k+ generated assets with a 75% positive feedback rate during a 6-week soft launch.
- Engineered a feature enabling users to add AI-generated/manual subtitles to audio/video content, sync and edit them using a **timeline-UI** made in **React**, and apply advanced styling/animations to them in a custom **Fabric.js** group.
- Scaled stock media offerings from 2,000 items in 2 categories to 300,000+ items across 6 categories by optimizing **MySQL** tables and relationships, automating optimization (using **ImageMagick** and **FFmpeg**) for presets stored on **AWS S3**, and implementing AI-based keyword/category tagging for efficient filtering and search with **AWS CloudSearch**.
- Reverse-engineered mathematical transformations used by **Fabric.js** item controls, to build custom div-based control elements enabling precise resizing, scaling, and rotation of canvas objects even if they were outside the canvas.
- Reduced image upload times by **60%** by implementing an **asynchronous** upload flow for generating required sizes, qualities, and variants in the background, with polling to handle errors.
- Devised and implemented an algorithm for instant color palette application and shuffling on items across complex canvas layouts, specifically catering to overlaps and z-index layering to ensure visual contrast.

PROJECTS

Lahore AQI Updates - Punjab Government EPA

Feb. 2025 – Present

Typescript, React Native, Node.js, Express, MongoDB, InfluxDB, Firebase

- Developed a **React Native Android App** with **Typescript**, delivering real-time and historical air quality data visualisations with interactive charts, 24-hour AQI forecasts, and personalized location-based push notifications.
- Handled state management using **React Context API**, custom hooks, AsyncStorage for persistent data with versioning, geolocation services, push notifications via **Firebase Cloud Messaging**, internationalization with **i18n**, device permissions management, and automated app updates.
- Built a scalable **Node.js + Express.js** backend that fetches the latest readings from 9 Lahore-wide EPA monitors, serves screen-specific data, and runs **cron** jobs to fetch daily weather information and run an ML model updating next-day AQI forecasts; using **Redis** for caching, semaphore locking, and session management.
- Integrated **MongoDB** to store user preferences, activity, and monitor JSON data; later migrated monitor data to **InfluxDB** when the government upgraded from hourly to minute-level readings, **reducing storage** costs by **30x** and **optimising** historical time-range queries by **200x**.

FactCheck: A Multilingual Web Scraper for Resource-Constrained Devices

June 2023 – Dec. 2023

AWS, HTML, CSS, JS, React.js, Flask, Web-Scrapping (beautifulSoup4), Cron, Caching, SSR

- Created an automated Python **web scraper** to fetch and translate fact-checked articles into Urdu, using a **cron** job to schedule Google Translate API calls and implemented **caching** with **Redis**, which reduced API costs by over 95%.
- Developed a web app (no longer hosted) to display the articles using React and Python (Flask), and optimized it for low-resource devices by implementing **Server-Side Rendering**, **lazy-loading**, and **infinite scroll**. Configured an **Apache** server on a **EC2** Ubuntu instance to deploy the webapp.

An Android Exploit for Audio Eavesdropping | React Native, Android Development

Jan 2023 – May 2023

- Developed an Android app with automated scripts to create Google Assistant-narrated audio files, record accelerometer data at customizable sampling frequencies, and convert the raw data into noise-free mel-spectrograms.
- Bypassed Android 12's 200 Hz sampling limit by exploiting sensor management vulnerabilities and developed a kernel-level interleaving **algorithm** to double IMU sampling frequency without user permissions.

YouTube AI Chatbot | React, Chrome APIs, Flask, manifest.json

March 2024

- Integrated **YouTube's Transcript Retrieval API** and **OpenAI's GPT API** to develop a **Chrome Extension** enabling users to query YouTube video content for detailed explanations via a chatbot.
- Incorporated **messaging services** between the content and background scripts, thus removing **2** unnecessary user actions (providing the video URL and sending an initiation message) found in several online examples.

TECHNICAL SKILLS

Languages and Frameworks: JavaScript/TypeScript, NodeJS, ReactJS, React Native, LangChain, LangGraph, RAG, AngularJS, Express, C/C++, Python, PHP, CodeIgniter, SQL, HTML/HTML5, CSS, JQuery

Developer Tools: Git, SVN, AWS (S3, EC2, CloudSearch, DynamoDB), MongoDB, InfluxDB, Google Cloud Platform, axe DevTools, WAVE, Docker, Storybook, Redux, Webpack, Expo, Cron, Bash Scripts, Apache/Nginx Server Configurations